Problem 4

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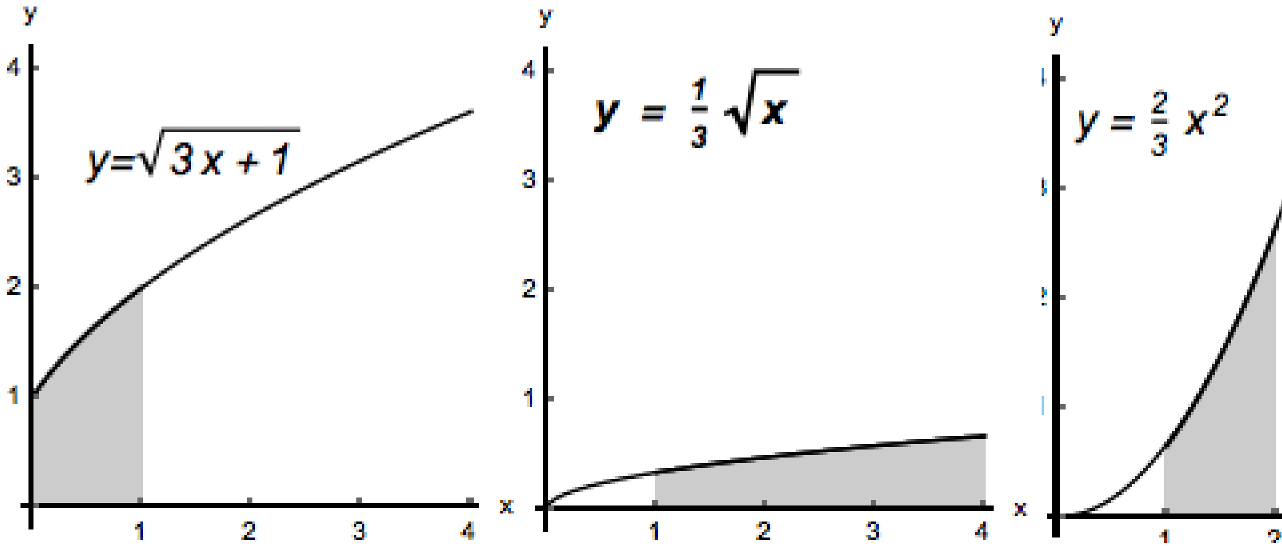
Problem 1 *Show that the areas of the three shaded regions in the figure below are equal. Find this area.*



Explanation. The area of the first shaded region is given by $\int_0^1 \sqrt{3x+1}\,dx$. We can evaluate this integral by substituting: $u = 3x + 1$, $du = 3\,dx$; when $x = 0$, $u = 1$; when $x = 1$, $u = 4$.

Therefore $\int_0^1 \sqrt{3x+1}\,dx = \int_1^4 \frac{1}{3}\sqrt{u}\,du$
and the last integral gives the area of the region in the middle.

The same integral can also be evaluated by substituting $u = \sqrt{3x+1}$, $du = \dfrac{3}{2\sqrt{3x+1}}\,dx = \dfrac{3}{2u}\,dx$;
when $x = 0$, $u = 1$; when $x = 1$, $u = 2$.

In this case we have
$\int_0^1 \sqrt{3x+1}\,dx = \int_1^2 u \cdot \frac{2}{3}u\,du = \int_1^2 \frac{2}{3}u^2\,du.$

The last integral gives the area of the third shaded region.
Therefore, the areas are equal.

In order to compute this area, let's evaluate the last integral.
$\int_1^2 \frac{2}{3}x^2\,dx = \frac{2}{3}\left[\frac{x^3}{3}\right]_1^2 = \frac{2}{3}\left[\frac{8-1}{3}\right] = \frac{14}{9}$